

# Mehak Negi

+91 8847678794 | [mehaknegi09@gmail.com](mailto:mehaknegi09@gmail.com) | [linkedin.com/in/mehak-negi](https://www.linkedin.com/in/mehak-negi) | [github.com/mehaknegi](https://github.com/mehaknegi)

## EDUCATION

### Chandigarh College of Engineering and Technology (CCET)

Bachelor of Engineering in Computer Science; GPA: 8.02/10.0

Chandigarh, India

Aug. 2023 – June 2027

## WORK EXPERIENCE & LEADERSHIP

### Technical Intern

Faramond Technologies

Summer 2025

Remote/On-site

- Engineered technical solutions for distributed systems, focusing on performance optimization and debugging.
- Researched microservices architecture to identify and mitigate latency bottlenecks in real-time environments.

### Database Head

Training and Placement Cell (TPC)

Jan. 2026 – Present

CCET

- Managing database structures and technical requirements for campus-wide technical events and placement drives.
- Leading a team of students to ensure data integrity and streamlined registration processes.

### Membership Chair

Association for Computing Machinery (ACM) Student Chapter

Jan. 2026 – Present

CCET

- Overseeing member enrollment and coordinating networking opportunities for the student community.
- Facilitating technical workshops and seminars to promote computing excellence for 100+ members.

## PROJECTS

### AI Chatbot for Mental Health | Python, TensorFlow, Keras, LSTM [Link](#)

2024

- Developed an LSTM-based RNN architecture for a mental health support chatbot, achieving **94.00% test accuracy**.
- Implemented an end-to-end NLP pipeline featuring tokenization, word embeddings, and categorical cross-entropy loss.
- Optimized response generation to provide context-aware emotional support through empathy-aware logic.

### Disney+ Hotstar Clone with Watch Party | React, Node.js, Socket.io, Firebase

2025

- Developed a full-stack video streaming application featuring a unique **Synchronized Watch Party** feature using **WebSockets**, allowing multiple users to sync video playback states with **100ms latency**.
- Engineered a custom video controller that handles play/pause/seek events across distributed clients in real-time.
- Optimized high-resolution UI rendering, reducing initial page load time by **30%** through lazy loading and asset compression.

## ACHIEVEMENTS & RESEARCH

### IEEE Research Publication | Latency Trade-Offs in Microservices (ICOIN)

2026

- Authored "A Novel Experimental Study on Latency Trade-Offs and Mitigation in Microservices Architectures" published at **ICOIN 2026** (DOI: 10.1109/ICOIN).
- Conducted comparative analysis of **gRPC, REST, and GraphQL** protocols under varying workloads to identify communication overheads.
- Proposed practical optimization strategies including **API Gateways, caching, and protocol-level tuning** to improve distributed system efficiency.

### GATE 2026 (Computer Science & IT) | Score: 35.69

March 2026

- Successfully qualified the National Level Graduate Aptitude Test in Engineering (GATE) on the **first attempt**, placing within the **top 15%** of candidates nationwide.
- Demonstrated proficiency in core Computer Science fundamentals including Data Structures, Algorithms, and Operating Systems at a competitive national scale.

## TECHNICAL SKILLS

**Languages:** C++, Python, MATLAB, SQL, JavaScript, HTML/CSS

**Libraries/Frameworks:** TensorFlow, Keras, OpenCV, Socket.io, React, Node.js

**Developer Tools:** Git, Docker, macOS, LaTeX, MATLAB Image Processing Toolbox

**Key Coursework:** Deep Learning, Operating Systems, Digital Image Processing, Microservices Architecture